

From Conservation Laws to the `TampinesSteamArray` Algorithm

A derivation of the 1-D compressible PIMPLE solver, its HEM closure,
and the three fixes that make the Edwards blowdown plateau appear

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AI-generated document. This derivation was generated with AI assistance from the conservation laws and from a reading of the `TampinesSteamArray` source. It is *explanatory background*, not a novel contribution: the finite-volume and PIMPLE material is standard textbook content (see the references), and is reproduced here only so that the project-specific parts at the end rest on a stated foundation.

It is written to be *re-derivable*. Every non-obvious step states *why* it is taken, because a derivation you cannot reconstruct is a derivation you do not have.

Statements about what the code does are tied to `src/openfoam_algorithms/rhoPimpleFoam/`. Where the implementation departs from stock `rhoPimpleFoam`, this is flagged explicitly in §11 rather than quietly idealised.

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1 The problem and the plan

We want a 1-D compressible pipe solver for water/steam that can survive a violent depressurisation (the Edwards–O’Brien blowdown), in which subcooled liquid flashes to a two-phase mixture and chokes at the break.

The chain of reasoning is:

1. Write the conservation laws (§2).
2. Discretise them by the finite-volume method, obtaining one algebraic equation per cell (§3).
3. Observe that pressure has *no equation of its own*, and manufacture one from continuity (§4).
4. Assemble those pieces into the PIMPLE loop (§5).
5. Close the system with IAPWS-IF97 steam tables instead of an ideal gas, which is where the difficulty lives (§6).
6. Repair the three failures that the steam-table closure exposes: energy conservation (§7), compressibility (§8), and shock capturing with its stability taper (§9).

2 Governing equations

For a compressible single-component fluid, with ρ density, $\mathbf{U}$ velocity, p pressure and h specific static enthalpy:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) = 0 \quad (1)$$

$$\frac{\partial (\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U} \mathbf{U}) = -\nabla p + \nabla \cdot (\mu \nabla \mathbf{U}) \quad (2)$$

$$\frac{\partial (\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{U} h) = \frac{\partial p}{\partial t} + \nabla \cdot (\alpha_h \nabla h) + Q \quad (3)$$

closed by an equation of state. For an ideal gas one writes $\rho = p/RT$; here we instead use the IAPWS-IF97 backward flash

$$\rho = \rho(p, h), \quad T = T(p, h), \quad \mu = \mu(p, h), \quad \alpha_h = \lambda(p, h)/c_p(p, h). \quad (4)$$

Unknowns versus equations — the orphan. Count them: the unknowns are $\mathbf{U}$, p , ρ , h . Momentum naturally yields $\mathbf{U}$; energy yields h ; the equation of state yields ρ . *Nothing yields p .* Pressure appears in (2) only as a gradient, and momentum is already committed to producing $\mathbf{U}$; one vector equation cannot deliver two fields.

What is left over is continuity, (1). So pressure is determined *implicitly*, by the requirement that the velocity field conserve mass. This single observation dictates the entire algorithm, and we return to it in §4.

3 Finite-volume discretisation

3.1 Integration and Gauss's theorem

Integrate each law over a control volume V_c and convert divergences to surface integrals:

$$\int_{V_c} \nabla \cdot \mathbf{F} dV = \oint_{\partial V_c} \mathbf{F} \cdot d\mathbf{S} \approx \sum_f \mathbf{F}_f \cdot \mathbf{S}_f, \quad (5)$$

where $\mathbf{S}_f$ is the *outward* face-area vector. The outward orientation is carried entirely by $\mathbf{S}_f$; it must not be applied a second time when approximating gradients.

Continuity (1) becomes, with mass flux $\phi_f = (\rho \mathbf{U})_f \cdot \mathbf{S}_f$ in kg/s,

$$V_c \frac{\rho_c - \rho_c^o}{\Delta t} + \sum_f \phi_f = 0. \quad (6)$$

Both terms are kg/s; omitting V_c or the dot product with $\mathbf{S}_f$ leaves the equation dimensionally inconsistent.

3.2 The diffusion stencil

For a diffusive flux the face gradient is a directional derivative in the outward direction, i.e. “ahead minus behind” with the neighbour ahead:

$$(\nabla \psi)_f \approx \frac{\psi_N - \psi_P}{\delta_f}, \quad \sum_f \Gamma_f |\mathbf{S}_f| \frac{\psi_N - \psi_P}{\delta_f}. \quad (7)$$

The sign is fixed by physics, independent of convention: if the neighbour is hotter or faster, cell P must *gain*. Expanding (7) gives one term in ψ_P and one in ψ_N .

3.3 The matrix that results

Every discretised transport equation collapses to one algebraic relation per cell, linking the cell to its neighbours:

$$a_P \psi_P + \sum_N a_N \psi_N = b. \quad (8)$$

Two distinctions must be kept apart, and conflating them is the classic error:

axis	question
implicit vs. explicit	is the value at the new time (unknown) or known?
diagonal vs. off-diagonal	is it <i>this</i> cell's value or a <i>neighbour's</i> ?

a_P is the coefficient on the cell's *own* unknown — the matrix diagonal. The a_N are off-diagonal. Both may be implicit. Deleting the off-diagonals would leave a purely diagonal matrix, invertible cell-by-cell, in which nothing diffuses: *the off-diagonal entries are the physical coupling*.

3.4 The H operator

Split the matrix $A = D + N$ into diagonal and off-diagonal parts. Then

$$(D + N)\mathbf{U} = b \implies D\mathbf{U} = b - N\mathbf{U} \implies \mathbf{U} = D^{-1}(b - N\mathbf{U}). \quad (9)$$

Writing $a_P \equiv D$ and $H(\mathbf{U}) \equiv b - N\mathbf{U}$, and separating the pressure gradient out of b :

$$\boxed{a_P \mathbf{U}_P = H(\mathbf{U}) - \nabla p} \quad (10)$$

So H is not an arbitrary bucket: it is precisely “the right-hand side after moving the off-diagonal action across.” It contains the neighbours’ contributions (still unknowns, at the new time) *plus* genuinely known sources such as the old-time term. It is written $H(\mathbf{U})$ because it depends on the velocity field and must be re-evaluated as the solution updates.

Equation (9) is a Jacobi/Gauss–Seidel splitting: the unknown appears on both sides. *That is why the scheme must iterate*, and it is the same structural reason the outer corrector loop exists.

3.5 Convection and its linearisation

Convection $\nabla \cdot (\rho \mathbf{U} \mathbf{U})$ is nonlinear — velocity transports itself. It is linearised by lagging the mass flux at the previous iterate while keeping the transported velocity implicit,

$$\sum_f \underbrace{\phi_f}_{\text{lagged coefficient}} \mathbf{U}_f^{\text{new}}, \quad (11)$$

which is a *Picard* (successive-substitution) linearisation. With first-order upwind, the face value is taken from the *upstream* cell:

$$\mathbf{U}_f = \begin{cases} \mathbf{U}_P, & \phi_f > 0 \quad (\text{outflow}) \Rightarrow \text{diagonal}, \\ \mathbf{U}_N, & \phi_f < 0 \quad (\text{inflow}) \Rightarrow \text{off-diagonal}. \end{cases} \quad (12)$$

Note that the lagged quantity is a *coefficient*, never the unknown: the velocity itself remains implicit.

4 Manufacturing the pressure equation

Divide (10) by a_P and define

$$r_{AU} \equiv \frac{1}{a_P}, \quad \text{HbyA} \equiv \frac{H(\mathbf{U})}{a_P}, \quad \text{so} \quad \mathbf{U}_P = \text{HbyA} - r_{AU} \nabla p. \quad (13)$$

HbyA is the velocity a cell would have from its neighbours and its own history alone — everything *except* the pressure force.

Now substitute (13) into discrete continuity (6). The only remaining unknown is p , and because ∇p sits inside a divergence we obtain a variable-coefficient Laplacian:

$$\sum_f (\rho r_{AU})_f (\nabla p)_f \cdot \mathbf{S}_f = V_c \frac{\rho - \rho^o}{\Delta t} + \sum_f \rho_f \text{HbyA}_f \cdot \mathbf{S}_f. \quad (14)$$

This is a *scalar* equation, matching the scalar unknown p . The orphan now has a home.

Reading the right-hand side. The term $\sum_f \rho_f \text{HbyA}_f \cdot \mathbf{S}_f$ is the mass imbalance of the pressure-free velocity. The pressure equation is therefore literally: *the size of the continuity violation tells you how to correct the pressure*.

Compressibility enters here. ρ on the right still depends on p . Linearising with

$$\psi \equiv \left. \frac{\partial \rho}{\partial p} \right|_h, \quad \rho \approx \rho^* + \psi (p - p^*), \quad (15)$$

and moving the resulting term to the left adds a diagonal contribution $\psi V_c / \Delta t$, making the equation Helmholtz-like rather than a pure Laplacian:

$$\boxed{\psi_c \frac{V_c (p_c - p_c^o)}{\Delta t} + \sum_f s_{c,f} \left[\phi_{\text{HbyA},f} - (\rho r_{AU})_f \text{snGrad}(p)_f |\mathbf{S}_f| \right] = 0} \quad (16)$$

In the incompressible limit $\psi \rightarrow 0$ and the classical pressure Poisson equation is recovered. Thus ψ sits *in the diagonal of the pressure equation*: it is the knob controlling how strongly pressure resists change. This will matter in §8.

5 The PIMPLE loop

Why a loop at all? Because the equations are coupled but we solve them *segregated*, one variable at a time. Each solve therefore uses stale values from the others, and the only repair is to go round again. A monolithic solver — all unknowns in one matrix — would need no predictor–corrector at all, but pays with one enormous, ill-conditioned system.

Two distinct lags are being repaired simultaneously:

lag	what is stale	example
nonlinearity (within an equation)	a coefficient depending on the unknown	lagged ϕ_f
coupling (between equations)	a value owned by another equation	∇p in momentum

The canonical ordering is forced, not chosen: momentum must be assembled first (it alone supplies the $\mathbf{U}$ – p relation), then continuity manufactures p , then velocity is corrected.

1. **Momentum predictor.** Solve (10) with the guessed p^* ; the resulting $\mathbf{U}^*$ does *not* conserve mass.
2. Form r_{AU} and $\mathbf{HbyA}$.
3. **Pressure corrector.** Solve (16).
4. **Velocity/flux correction** with the new p ; continuity now holds.
5. Repeat 3–4 (PISO inner correctors); refresh and repeat all (PIMPLE outer).

SIMPLE, PISO and PIMPLE differ only in how many times one goes round and what is refreshed. PIMPLE contains both as limits: one outer corrector reduces it to PISO; dropping the time derivative and relaxing heavily gives SIMPLE.

6 Closure by steam tables, and why it breaks things

Replacing the ideal gas by the IAPWS-IF97 flash (4) is not a cosmetic substitution. Inside the saturation dome:

- ρ responds violently to p (a small pressure drop flashes liquid to vapour), so ψ is orders of magnitude larger than any single-phase value;
- the equilibrium sound speed collapses, so the flow becomes near-sonic at modest velocities;
- the flash has hard validity limits — a state pushed below 273.15 K or above 1073.15 K does not merely lose accuracy, it *panics*.

Each of the three fixes below is a direct consequence of one of these facts.

7 Fix 1 — conservative energy and the continuity density

7.1 Why the naive form leaks

In continuous calculus the conservative and non-conservative energy forms agree:

$$\begin{aligned} \frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{U} h) &= \rho \frac{\partial h}{\partial t} + h \frac{\partial \rho}{\partial t} + \rho \mathbf{U} \cdot \nabla h + h \nabla \cdot (\rho \mathbf{U}) \\ &= \rho \frac{Dh}{Dt} + h \underbrace{\left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) \right]}_{\text{continuity}}. \end{aligned} \quad (17)$$

The bracket vanishes *only because continuity holds exactly*. If continuity is satisfied to within a residual r , the two forms differ by exactly $h r$ — a spurious energy source or sink.

Discretely r is generally non-zero, because the thermodynamic density $\rho(p, h)$ returned by the flash and the density implied by the discrete mass balance are computed by *different parts of the solver* and need not agree. During flashing, ρ swings enormously and the leak $h r$ becomes large: enthalpy is over-drained, the liquid is driven subcooled, and the pressure falls straight past p_{sat} instead of holding the flashing plateau.

7.2 The continuity-consistent density

Rearranging discrete continuity (6) for the new density defines

$$\boxed{\rho_{\text{cont}} \equiv \rho^o - \frac{\Delta t}{V_c} \sum_f \phi_f = \rho^o - \Delta t \nabla \cdot \phi} \quad (18)$$

By construction $(\rho_{\text{cont}} - \rho^o)/\Delta t = -\nabla \cdot \phi$ holds *exactly*, whatever the equation of state returns. Using ρ_{cont} in the storage term therefore makes the cancellation in (17) exact, term for term, and the energy equation reduces to the material derivative $\rho Dh/Dt = \partial p/\partial t$. That reversible $dh \approx dp/\rho$ is what keeps the state on the saturation dome as p falls — i.e. *the plateau*.

As implemented. The energy time derivative is assembled by `fvm::ddt_coeff_old`, giving

$$\frac{\rho_{\text{cont}} h - \rho^o h^o}{\Delta t} V_c, \quad (19)$$

with two coupled points: the old-time coefficient is the *old* density ρ^o (restoring the missing $h^o(\rho - \rho^o)/\Delta t$ term), and the new-time coefficient is ρ_{cont} recomputed from the *final* mass flux — not the EOS density that the thermodynamic update wrote during the inner correctors. Using the EOS density instead leaves a residual that spuriously *over-heats* cells, flashing them into Region 5.

ρ_{cont} is floored at $1 \times 10^{-4} \text{ kg/m}^3$. In a clamped cell the exact cancellation formally no longer holds — a caveat that returns in §9.

8 Fix 2 — the compressibility ψ

With energy conserved, the flash initiates at the correct pressure but *overshoots* the plateau. The fault is now in the pressure equation.

A quality-frozen isothermal compressibility $\psi = \rho \kappa_T$ omits the flashing contribution $(v_g - v_f) dx/dp$ and is roughly two orders of magnitude too small inside the dome. Since ψ occupies the diagonal of (16), too small a ψ makes the solver behave as though the fluid were nearly incompressible ($\psi \rightarrow 0$ is exactly the incompressible limit), so it fails to resist the depressurisation and sails through the plateau.

The remedy is to evaluate the true derivative along the thermodynamically relevant path — constant enthalpy, since h is the transported variable and a fast blowdown is approximately adiabatic — by central difference on the real flash:

$$\psi = \frac{\rho(p + \Delta p, h) - \rho(p - \Delta p, h)}{2 \Delta p}, \quad \Delta p = \max(10^{-3}p, 50 \text{ Pa}). \quad (20)$$

The bounds are clamped to the EOS validity range and the *actual* ($p_{\text{hi}} - p_{\text{lo}}$) is used as the denominator, so clamping degrades the estimate to a one-sided difference rather than corrupting it. If both bounds collapse together the isothermal value is used as a fallback, and ψ is floored at 10^{-12} .

With (20) the plateau emerges from the thermodynamics alone — no clamps, no saturation clips.

Cost. Two extra (p, h) flashes per cell per thermodynamic update, and the update runs once per inner corrector: $2 n_{\text{inner}} n_{\text{outer}} n_{\text{cells}}$ extra flashes per timestep.

9 Fix 3 — all-Mach shock capturing and its taper

9.1 Why dissipation is needed

In this solver the energy convection term is assembled *explicitly with central interpolation*. Central differencing of convection is unconditionally non-monotone: at the near-sonic flashing front it rings. A pressure-based solver has almost no numerical dissipation of its own, so the oscillation is not damped.

9.2 KNP central-upwind flux

Borrow the Kurganov–Noelle–Petrova central-upwind flux from density-based schemes. With one-sided local wave-speed estimates built from the *HEM equilibrium* sound speed c ,

$$a_L = \min(u_{n,L} - c_L, u_{n,R} - c_R, 0), \quad a_R = \max(u_{n,L} + c_L, u_{n,R} + c_R, 0), \quad (21)$$

the flux of a conserved variable w is

$$F = \frac{a_R F_L - a_L F_R + a_L a_R (w_R - w_L)}{a_R - a_L}. \quad (22)$$

The final term is the dissipation. Subtracting the corresponding central flux (the same expression with the jump term removed) isolates it, and it is applied as a *deferred correction* added to ϕ rather than by rebuilding the matrix. Left and right states are MUSCL-reconstructed with a van Leer limiter.

9.3 Three non-obvious choices

(a) Dissipate ρh_e , not ρE . The conserved energy variable is the *static* enthalpy density. Dissipating total energy $\rho E = \rho(h_e + \frac{1}{2}|\mathbf{U}|^2) - p$ would re-inject the $-\Delta p$ pressure work; across the strong rarefaction the large Δp then over-cools the near-break cell through the 273.15 K validity edge.

(b) Gate on $\min(\text{Ma})$, not $\max$. The blend weight is

$$\beta = \text{clamp}\left(\frac{\text{Ma}_f - \text{Ma}_{\text{lo}}}{\text{Ma}_{\text{hi}} - \text{Ma}_{\text{lo}}}, 0, 1\right), \quad \text{Ma}_f = \min(\text{Ma}_O, \text{Ma}_N), \quad (23)$$

with defaults $\text{Ma}_{\text{lo}} = 0.3$, $\text{Ma}_{\text{hi}} = 1.0$. At a liquid/two-phase interface the liquid's ~ 1400 m/s sound speed dominates the wave speeds, so the numerical viscosity $a_L a_R / (a_R - a_L)$ is enormous — yet that liquid acoustic wave is genuinely *low Mach* and must not be dissipated. Taking the minimum sees the subsonic liquid side and returns $\beta = 0$; only where *both* sides are near-sonic does KNP activate.

(c) No separate energy source. The continuity dissipation is folded into ϕ *before* the energy equation is assembled, so $\nabla \cdot (\phi h)$ already carries it. Adding a standalone ΔF_{ener} would double-count.

9.4 The rarefied-tail taper

Even so, the full 600 ms run failed at $t \approx 0.18$ s. As the pipe empties, ρ_{cont} falls to its floor and the energy-equation diagonal $\rho_{\text{cont}} V_c / \Delta t$ collapses; the cell becomes ill-conditioned, and a fixed explicit correction then produces an unbounded enthalpy update that trips the (p, h) validity edge.

Since the minimum dissipated-face density during the physics of interest is ≈ 106.5 kg/m³, a density threshold cleanly separates the runaway from the region that matters. The blend weight is tapered:

$$g(\rho_f) = \text{clamp}\left(\frac{\rho_f - 50}{100 - 50}, 0, 1\right), \quad \rho_f = \min(\rho_O, \rho_N), \quad \beta \leftarrow \beta g(\rho_f) \quad (24)$$

Below 50 kg/m³ the dissipation is switched off entirely (pure PIMPLE, stable); above 100 kg/m³ it is at full weight, so the taper is *inert* in the validated region and the plateau is unchanged. The physical justification is that in the emptying tail there is no flashing shock left to capture.

The effective face weight is therefore

$$\beta_f = \text{clamp}\left(\frac{\min(\text{Ma}_O, \text{Ma}_N) - 0.3}{0.7}, 0, 1\right) \cdot \text{clamp}\left(\frac{\min(\rho_O, \rho_N) - 50}{50}, 0, 1\right), \quad (25)$$

further gated by a validity guard requiring both cells to lie within 300 K to 1050 K.

10 The assembled algorithm

Per timestep, with u^o, p^o, h^o, ρ^o the previous state:

1. **Outer loop**, n_{outer} times:

- (a) **Continuity (explicit).** $\rho = \max(\rho^o - \Delta t \nabla \cdot \phi, 10^{-4})$.
- (b) **Assemble momentum:** implicit Euler time term, first-order implicit upwind convection, orthogonal Laplacian diffusion; add the lagged hybrid momentum source if in `HybridAllMach` mode.
- (c) Cache $A = \text{diag}$ and $r_{AU} = V_c / A$.
- (d) **Momentum predictor:** add $-V_c \nabla p$ explicitly, solve for $\mathbf{U}$, then *remove* it again so that $H(\mathbf{U})$ is pressure-free for the projection.
- (e) **Inner loop**, n_{inner} times: form HbyA and ϕ_{HbyA} ; assemble and solve (16); under-relax and bound p ; correct the internal-face fluxes $\phi = \phi_{\text{HbyA}} - (\rho r_{AU})_f \text{snGrad}(p) |\mathbf{S}_f|$; correct $\mathbf{U} = \text{HbyA} - r_{AU} \nabla p$; re-close the thermodynamics via the (p, h) flash (which also refreshes ψ by (20)).
- (f) **Hybrid dissipation** (if enabled): fold $\Delta \phi$ into ϕ immediately; store the momentum source for the *next* outer corrector.

- (g) **Energy:** build ρ_{cont} from the final ϕ , assemble the conservative ddt with $(\rho_{\text{cont}}, \rho^o)$, explicit central convection, implicit α_h diffusion, $\partial p / \partial t$ source, plus lateral conductance and volumetric heat sources; solve for h .

- 2. Clear the per-step lateral/heat registrations.

Boundary conditions. A closed end is a Dirichlet $\mathbf{U} = 0$. The break is a *choked-flow outlet*: the critical pressure and mass flux are obtained from the HEM critical-flow dispatcher using the last cell's (p_0, h_0) , converted to an equivalent full-face velocity so that $\dot{m}$ is conserved through the smaller break area, and imposed as a Dirichlet velocity (lagged one step). Both pressure patches remain zero-gradient: a blowdown has no Dirichlet pressure anywhere — mass depletion sets p .

11 Deviations from stock rhoPimpleFoam

These are deliberate or known, and are recorded so the document is not read as describing textbook behaviour.

1. **Equation order.** Stock is $\text{rhoEqn} \rightarrow \text{UEqn} \rightarrow \text{EEqn} \rightarrow \text{pEqn}$; here energy is solved *after* the pressure loop, so h lags p by one outer corrector.
2. rhoEqn is recomputed every outer corrector, not only on the first iteration.
3. **Energy convection is explicit and central**, not implicit limited upwind. This is the dispersion the KNP hybrid exists to control.
4. No ddtCorr (Rhie–Chow transient flux correction) in ϕ_{HbyA} .
5. No kinetic-energy or gravity-work terms in the energy equation, and no viscous dissipation.
6. The *momentum* time derivative is non-conservative $(\rho^{\text{new}}(\mathbf{U} - \mathbf{U}^o) / \Delta t)$ while the *energy* time derivative is conservative. Asymmetric by design.
7. No residual-based PIMPLE convergence test: corrector counts are fixed and solver performance is discarded. Only the linear-solver tolerances apply.
8. Hybrid *momentum* dissipation is lagged one outer corrector, so with $n_{\text{outer}} = 1$ it is computed and then discarded. The Edwards runs use $n_{\text{outer}} = 4$.
9. Hybrid dissipation and the flux correction touch internal faces only.
10. Numerical floors $(\rho, \rho_{\text{cont}} \geq 10^{-4}, \psi \geq 10^{-12}, c \geq 1 \text{ m/s})$ formally break the exact discrete-continuity cancellation in clamped cells.
11. Enthalpy boundary conditions are not re-stamped after a solve, so a Dirichlet enthalpy BC survives one step and then silently degrades to zero-gradient. This does not affect Edwards, which sets no enthalpy BC.
12. No turbulence model, wall function, wall friction, mesh motion, LTS, or transonic pcEqn variant.

12 Summary

The three project-specific fixes are one idea in three costumes: *keep the discrete system self-consistent*.

fix	failure it removes	mechanism
$\rho_{\text{cont}} + \text{conservative ddt}$	plateau collapses to 17.4 psia	exact discrete-continuity cancellation
$\psi = \partial\rho/\partial p _h$	plateau overshoots	correct stiffness in the pEqn diagonal
KNP + min(Ma) + taper	ringing; tail crash at 0.18 s	selective, gracefully deactivated dissipation

None of them is a tuned fudge: each is the consequence of requiring that the discrete equations mean what the continuous ones mean.

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